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A novel hybrid particle swarm optimization for multi-UAV cooperate path planning

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Abstract

The path planning of unmanned aerial vehicle (UAV) in three-dimensional (3D) environment is an important part of the entire UAV's autonomous control system. In the constrained mission environment, planning optimal paths for multiple UAVs is a challenging problem. To solve this problem, the time stamp segmentation (TSS) model is adopted to simplify the handling of coordination cost of UAVs, and then a novel hybrid algorithm called HIPSO-MSOS is proposed by combining improved particle swarm optimization (IPSO) and modified symbiotic organisms search (MSOS). The exploration and exploitation abilities are combined efficiently, which brings good performance to the proposed algorithm. The cubic B-spline curve is used to smooth the generated path so that the planned path is flyable for UAV. To assess performance, the simulation is carried out in the virtual three-dimensional complex terrain environment. The experimental results show that the HIPSO-MSOS algorithm can successfully generate feasible and effective paths for each UAV, and its performance is superior to the other five algorithms, namely PSO, Firefly, DE, MSOS and HSGWO-MSOS algorithms in terms of accuracy, convergence speed, stability and robustness. Moreover, HIPSO-MSOS performs better than other tested methods in multi-objective optimization problems. Thus, the HIPSO-MSOS algorithm is a feasible and reliable alternative for some difficult and practical problems.

Keywords Unmanned aerial vehicle (UAV) · Cooperate path planning · Particle swarm optimization · Time stamp segmentation (TSS) model · Symbiotic organisms search (SOS)

1 Introduction

In the last decades, unmanned aerial vehicle (UAV) has attracted more attention for its ability to work efficiently in complex and dangerous environments. Multi-UAV path planning problem is one of the challenging tasks in the field of UAVs for the researcher. The main objective of the problem is to find a collision free optimal flight path between the source and destination without colliding with any of the colleagues or obstacles present in an environment, while also meeting the kinematic constraints of UAV to ensure that the planned path is flyable. To meet the above objectives, the flight path planning problem can be formulated as a multi-objective optimization problem.

The existing path planning methods are mainly divided into two categories: traditional methods and heuristic methods. Outstanding traditional methods include A* algorithm [1, 2], potential field method [3, 4], Voronoi-diagram [5] and visibility graph-based [6]. Although the traditional method is simple to implement, it lacks flexibility and can only be applied to simple task scenarios. Therefore, heuristic algorithms have become more popular to solve complex optimization problem [7, 8]. Heuristic planning methods mainly include ant colony optimization (ACO) [9, 10], particle swarm optimization (PSO) [11–13], firefly algorithm (FA) [14, 15], differential evolution (DE) [16, 17], genetic algorithm (GA) [18–20]. Heuristic algorithms can be used to solve NP-Hard problems and obtain near-optimal solutions, while path planning problems are actually a type of NP-Hard problems. However, most heuristic algorithms have the disadvantages of slow convergence and local optimum. So far, there is not any

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method that can completely solve UAV path planning problem. Therefore, it is necessary to seek new and more effective solutions or improve existing methods.

Researchers have tried to apply various intelligent algorithms to solve the path planning problem of robots or UAVs, and have made good progress. For instance, the authors of [21] used a heuristic evolution algorithm to find the optimal flight path. The authors of [22] presents an evolutionary algorithm-based UAV path planner based on a novel separate evolution strategy. A novel phase angle-encoded fruit fly optimization algorithm with mutation adaptation mechanism is presented for UAV path planning [23]. In [24], an enhanced simulated annealing (ESA) approach was proposed for robot path planning in dynamic environments with both static and dynamic obstacles. For multiple robots path planning, an on-line collision-free path planning strategy was developed by using the proposed artificial bee colony (EABC) algorithm in [25], a co-evolutionary improved genetic algorithm (CIGA) was used in [26]. In order to overcome the shortcoming of single heuristic algorithm, some researchers adopted a new kind of technique called hybridization [27, 28], which effectively combines multiple algorithms together and has achieved good results. The hybrid optimization approach proposed in [29] combines the classification characteristics of a clustering method with the randomness of chaotic sequences to examines the search space efficiently. To solve the multi-UAV path planning problem, the authors of [30] developed a modified central force optimization (MCFO) algorithm, which introduces particle swarm optimization ideas and genetic algorithm mutations into the CFO. The original wolf pack search (WPS) algorithm is modified by the crossover operator and mutation operator of genetic algorithm to acquire a great convergence [31]. By combining the simplified gray wolf optimization (SGWO) and the modified SOS (MSOS), a new hybrid algorithm called HSGWO-MSOS is proposed to improve exploitation abilities [32].

Among the existing intelligent algorithms, particle swarm optimization is widely used to solve various optimization problems due to its flexibility, simplicity and ease of implementation. Many studies have been carried out in recent years, which attempt to solve the path planning problem by using PSO algorithms. [33] developed a PSO-based algorithm, called dPSO-U, for exploring a disaster scenario area. A smooth path planning of a mobile robot has solved using Stochastic PSO [34]. The authors in [35] proposed a hybrid algorithm based on PSO and metropolis criterion to reduce the probability of the PSO falling into a local optimum. The authors in [36] improved the PSO and the gravitational search algorithm (GSA), and the proposed hybridization IPSO-IGSA maintain the efficient balance between exploration and exploitation. The author in [37] using hybridization of improved PSO (IPSO) with differentially perturbed velocity (DV) algorithm to improve the exploration and exploitation in IPSO-DV.

However, PSO suffers from premature convergence in the evolutionary process while dealing with path planning problems. Although some existing studies have improved PSO and many variants with better performance have been proposed, there are still some shortcomings. The existing researches on applying PSO to path planning has the following deficiencies: (1) As more control parameters that need to be tuned by the user are added, PSO becomes complicated. (2) Hard to maintain the efficient balance between exploration and exploitation. (3) Most studies only consider the two-dimensional environment, ignoring the UAV's kinematic constraints and coordination between multi-UAVs, which makes the planned path unflyable for UAV. Therefore, this paper improves the PSO in response to these deficiencies, and applies the improved PSO to the path planning of multiple UAVs in a complex 3D environment.

Cheng and Prayogo [38], inspired by the symbiotic interaction between organisms, proposed a new algorithm called symbiotic organisms search (SOS) in 2014. In SOS, current organism produces offspring through symbiotic behavior with other random or best organism. This method has great exploitation capabilities. Because the SOS algorithm does not need to adjust parameters and is easy to implement, it has been widely used. In this paper, a novel hybrid particle swarm optimization algorithm named HIPSO-MSOS is proposed to solve the multi-UAV path planning problem, in which the exploration and exploitation capabilities of the PSO and SOS algorithms are effectively combined. In the proposed algorithm, the PSO is improved by three effective strategies: (1)The metropolis criterion can effectively prevent PSO from falling into a local optimum. (2) The linear change strategy of the weight coefficients in the algorithm is conducive to balancing the global exploration and local exploitation capabilities. (3) Instant update strategy provides the latest information about the solution. In addition, the SOS algorithm was modified to improve its exploitation capability.

The main objective of this paper may be summarized as follows: (i) we study the problem of multi-UAV path planning in a clutter environment and formulated the above problem as multi-objective optimization problem with constraints; (ii) Three effective strategies are used to improve PSO, and then by combining the improved PSO with the modified SOS algorithm, a novel method to the solution of optimal path generation for multi-UAV path planning problem using HIPSO-MSOS is proposed in this paper; (iii) The proposed algorithm has been applied to multi-UAV path planning in complex environment, and the obtained results are compared to other optimization algorithms such as PSO, DE.

For solving multi-UAV cooperate path planning problem with hybridization of IPSO-MSOS, the multi-UAV path planning model is first constructed by using TSS technology [39], which can simplify the handling of coordination cost of UAVs and reduce the dimension of solution space. Then, in order to evaluate the planned path, the cost function is designed based on the length, height, smoothness of the path and the penalty from the constraints. By combining the improved PSO with the modified SOS algorithm, a hybrid algorithm HIPSO-MSOS is proposed, which maintain the efficient balance between exploration and exploitation capability. A particle in HIPSO-MSOS algorithm is a solution in the solution space of the optimization problem. The purpose of the algorithm is to provide a strategy to find the optimal solution in this solution space. Finally, the cubic B-spline curve is used to smooth the generated path so that the planned path is flyable for the UAV. The result shows that the algorithm can improve the solution quality and the optimal paths are obtained with the help of HIPSO-MSOS algorithm.

The rest of the paper is organized as follows: In Section 2, the TSS model is introduced and the cost function of the multi-UAV path planning problem is designed. The principles of the basic PSO and SOS algorithm are explained in Section 3. Section 4 describes the proposed HIPSO-MSOS algorithm in detail and the path smoothing method by the cubic B-spline curve is introduced in section 5. The comparative experiment is completed in Section 6. Finally, Section 7 gives the conclusions.

2 Problem formulation

2.1 Overview of path planning

Multi-UAV cooperative path planning is to generate path for each UAV to meet the requirements of safety, short-range and cooperation. Safety describes the need to avoid obstacles and hazardous areas in complex environments. Short-range means less fuel and energy costs. Cooperation is determined by the mission requirements of the UAV swarm. For example, an attack mission requires the UAV to reach the specified target location at the same time.

Generally speaking, the path planning probably includes environmental modeling, determining the objective function to be optimized, selecting the appropriate solution strategy, and path smoothing, as shown in Fig. 1.

Here are the steps for the UAV path planning.

Step 1: Determine the planning space for path planning based on the starting point, task area, environmental terrain conditions, and threat distribution;

Step 2: Determine the constraints and mathematical models of path planning based on task constraints and aircraft performance limitations;

Step 3: Select the appropriate search algorithm to find the optimal feasible path that meets the constraints;

Step 4: Smooth the planned path;

Step 5: Output planned paths;

Specifically, the resulting target trajectories must satisfy the following constraints:

(1) Maximum yaw angle: the UAV can only turn within the maximum yaw angle determined by its horizontal plane maneuverability.

(2) Maximum pitch angle: the UAV can only turn within the maximum pitch angle determined by its vertical plane maneuverability.

(3) Maximum and minimum flight altitude: in order to ensure the safety of UAV, the flight altitude of UAV shall be greater than the minimum altitude pre-determined and less than the maximum altitude pre-determined.

(4) No collision: considering the size of UAV, when UAV is flying along the planned path, it cannot collide with the terrain, obstacles or colleagues in the scene.

Only the path that meets the above constraints is the feasible path. In addition, on the premise that the path is feasible, the path must be optimized as much as possible. The objective of optimization is to make the path short, the height fluctuation small, turns less and avoid the detection of radar.

2.2 Time stamp segmentation model

The TSS model can provide common time reference for path planning of multiple UAVs, so as to simplify the calculation of path coordination cost.

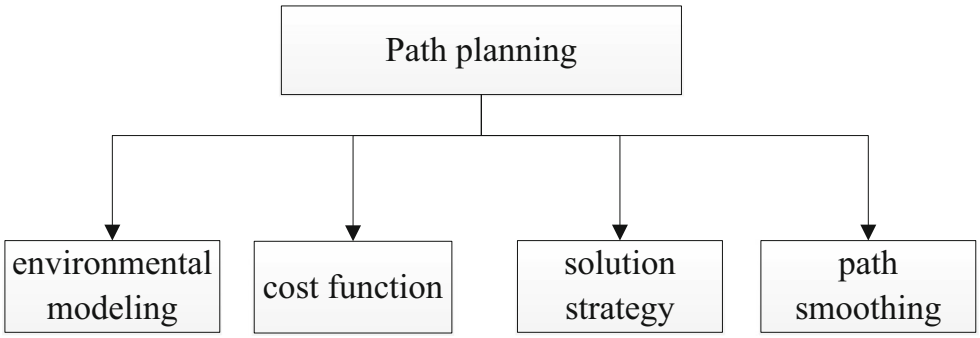
Assume that all the UAVs share the same takeoff time T_t and arrival time T_a. Given the number of path point D, the time stamp t_s is defined as $t_s = (T_a - T_t)/D$, where t_s is calculated by (2). In the TSS model, two components namely the velocity component $V_{m,n} = [V_{m,n,x}, V_{m,n,y}]$ and position component $P_{m,n} = [P_{m,n,x}, P_{m,n,y}]$ are designated, where $V_{m,n,x}$ and $V_{m,n,y}$ denote the velocity in the x-axis and y-axis directions, respectively. $P_{m,n}$ is updated according to $V_{m,n}$ by

$$P_{m,n} = P_{m,n-1} + V_{m,n} \cdot t_s \tag{1}$$

where $V_{m,n}$ and $P_{m,n}$ are the horizontal speed and horizontal position of the mth UAV at the nth timestamp, respectively.

In the TSS model, the range of the velocity component and the time stamp are determined by the in-flight

Fig. 1 path planning module



constraints of UAVs such as the range of airspeed $[V_{min}, V_{max}]$. Suppose that straight-line distance between the takeoff point and target point of mth UAV is l_m, the range of time stamp t_s is

$$t_s \in \bigcap_{m=1}^{M} [t_{m,\min}, t_{m,\max}],$$
$$t_{m,\min} = \frac{l_m}{V_{\max} D}$$
$$t_{m,\max} = \frac{l_m}{V_{\min} D} \tag{2}$$

where M is the number of UAV and $\cap$ denotes the intersection. After selecting the time stamp t_s, we can easily get the common arrival time $T_a = Dt_s + T_t$.

Assuming that the UAV flight altitude H is not considered first, the starting point O_m of the UAV points to the end point E_m as the x axis of the UAV's local coordinate system $X_m O_m Y_m$. The local coordinate system of UAV is shown in Fig. 2. In the local coordinate system, the velocity of UAV in the x-axis direction is constant, that is $\overline{V}_{m,n,x} = l_m/(Dt_s)$.

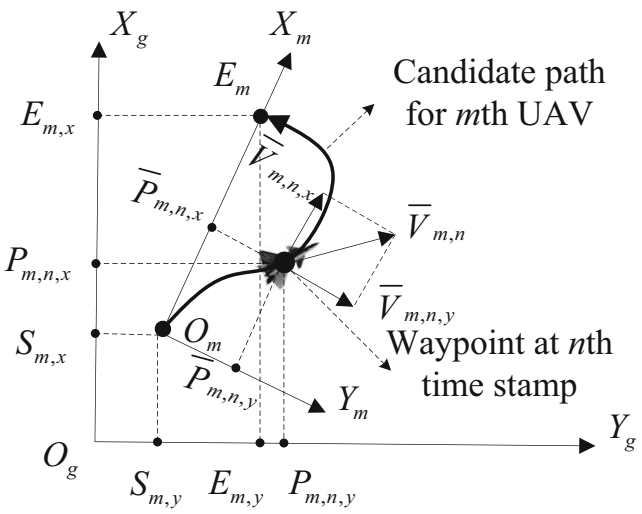


Fig. 2 TSS path planning model

Then use the optimization algorithm to search for the optimal configuration of the height components and the velocity components in the y-axis direction $\overline{V}_{m,n,y}$ at each path point. After that, according to the determined time stamp t_s, the coordinates $\overline{P}_{m,n} = [\overline{P}_{m,n,x}, \overline{P}_{m,n,y}]$ of the path point in the local coordinate system can be determined. Then, according to $\overline{P}_{m,n}$ and the corresponding height H, the 3D coordinates of the path points in the global coordinate system can be obtained, so as to obtain the required path. Considering the maximum yaw angle ψ_{max} and the velocity component in the x-axis direction $\overline{V}_{m,n,x}$, we can determine the range of velocity components in the y-axis direction, that is $\overline{V}_{m,n,y} \in [-\overline{V}_{m,n,x} tan\psi_{max}, \overline{V}_{m,n,x} tan\psi_{max}]$, $m = 1 \ldots M, n = 1 \ldots D$. Similarly, considering the maximum pitch angle ϕ_{max} and horizontal velocity , the range of the height difference between adjacent path points is $\Delta H_{m,n} \in [-\overline{V}_{m,n} t_s tan\phi_{max}, \overline{V}_{m,n} t_s tan\phi_{max}]$. Therefore, the search space is T-dimensional, where $T = M * D * 2$, that is $S_{Search} = \big(\overline{V}_{1,1,y}, \Delta H_{1,1}, \ldots,$
$\overline{V}_{1,D,y}, \Delta H_{1,D}, \ldots, \overline{V}_{2,1,y}, \Delta H_{2,1}, \ldots, \overline{V}_{M,D,y}, \Delta H_{M,D}\big)$.

2.3 Cost function

The principle of cost function design is mainly to consider what kind of requirements the planned path should meet. The performance goals to be optimized considered in this paper are

1) The path should be as short as possible, so that the energy consumption is small and the task time is short.
2) The path turning angle is small, which is also conducive to energy saving.
3) The path height is as low as possible and the change is small, which is conducive to energy saving and improved safety.
4) When UAV is flying along the path, it is avoided to be covered by the radar's detection range, otherwise it will be threatened.

5) The UAV should avoid collision with terrain, obstacles or other UAVs when flying along the path, otherwise the planning fails.
6) The UAV should be able to meet its own kinematical constraints when flying along the path.

According to the above requirements, the target performance that needs to be achieved is represented by different costs.

(1) Path length costs

Since the path is composed of a series of path points, the path length cost of mth UAV is defined as

$$J_m^{length} = \sum_{\mathrm{k}=1}^{D} \|(x_{k+1}, y_{k+1}, z_{k+1}) - (x_k, y_k, z_k)\|_2 \tag{3}$$

Where (x_k, y_k, z_k) is the coordinate of the kth path point in the path and D represents the number of path points (not include the end point).

(2) Path smoothing costs

The sum of the angles between the path segments is taken as the smoothing cost. The path smoothing cost of mth UAV is expressed as

$$J_m^{smooth} = \sum_{k=1}^{D-1} \arccos \frac{\varphi_{k+1}\varphi_k}{|\varphi_{k+1}||\varphi_k|} \tag{4}$$

where the vector φ_k is $(x_{k+1}-x_k, y_{k+1}-y_k, z_{k+1}-z_k)$.

(3) Path altitude costs

The UAV has low flight altitude and small altitude change, so it has low energy consumption and high safety. The path altitude cost of mth UAV is presented as

$$J_m^{altitude} = h_1\overline{y} + h_2\frac{1}{n}\sum_{k=1}^{n}(y_k-\overline{y})^2$$
$$\overline{y} = \frac{1}{D}\sum_{k=1}^{n} y_k \tag{5}$$

where h_1 and h_2 are the weight of path average altitude and the weight of altitude variance, respectively. y_k is the height of the kth path point.

(4) Threat costs from radar

If the distance between any UAV and the radar center is less than the detection radius of the radar, the UAV will be threatened, resulting in a threat cost. The threat level depends on the number of UAVs covered by the radar coverage and the distance between these UAVs and the radar center. Then the threat cost of mth UAV from the radar is defined as:

$$J_m^{Radar} = \sum_{k=1}^{D}\sum_{r=1}^{R} J_{m,k,r}^{Radar}$$
$$J_{m,k,r}^{Radar} = \begin{cases} \frac{\lambda}{\|P_{m,k}-P_r\|} & , \|P_{m,k}-P_r\| \le R_r \\ 0 & , otherwise \end{cases} \tag{6}$$

where $J_{m,k,r}^{Radar}$ is the cost of the mth UAV being threatened by the rth radar at the kth waypoint. $P_{m,k}$ represents the three-dimensional position of the m th UAV at the kth point, P_r is the location of the rth radar center, λ is the threat coefficient of the radar, and R_r is the detection radius of the rth radar.

(5) Threat costs from terrain

When the UAV collides with the terrain while flying along the path, the resulting terrain threat Q is extremely costly. The cost of the mth UAV being threatened by terrain is expressed as

$$J_m^{Terrain} = \sum_{k=1}^{D} J_{m,k}^{Terrain}$$
$$J_{m,k}^{Terrain} = \begin{cases} Q & , uav_m \text{collide with terrain at} k\text{th pathpoint} \\ 0 & , \text{otherwise} \end{cases} \tag{7}$$

(6) Threat costs from obstacles

When the UAV collides with the obstacles while flying along the path, the resulting obstacles threat Q is extremely costly, that is

$$J_m^{Obstacle} = \sum_{k=1}^{D}\sum_{o=1}^{O} J_{m,k,o}^{Obstacle}$$
$$J_{m,k,o}^{Obstacle} = \begin{cases} Q & , uav_m collide with o\text{th obstacle} \\ 0 & , \text{otherwise} \end{cases} \tag{8}$$

where O denotes the number of obstacle and $J_{m,k,o}^{Obstacle}$ is the cost of the mth UAV being threatened by the oth obstacle at the kth waypoint.

(7) Coordination costs

When the UAV fly along the path, they can't collide with each other, otherwise the coordination cost is enormous. The cost of coordination between UAVs is expressed as

$$J^{Coor} = \sum_{i=1}^{M}\sum_{j=i+1}^{M}\sum_{k=1}^{D} J_{i,j,k}^{Coor}$$
$$J_{i,j,k}^{Coor} = \begin{cases} Q & , \|P_{i,k}-P_{j,k}\| \le \delta_{uav} \\ 0 & , \text{otherwise} \end{cases} \tag{9}$$

where $J_{i,j,k}^{Coor}$ represents the coordination cost of the ith UAV and the jth UAV at the k th waypoint, $P_{i,k}$ is the position of the ith UAV at the k th waypoint, δ_{uav} is the safe distance between UAVs and M is the number of UAV.

In summary, the cost function is defined as:

$$\begin{aligned} J_{cost} =& w_1 \sum_{m=1}^{M} J_m^{length} + w_2 \sum_{m=1}^{M} J_m^{smooth} + w_3 \sum_{m=1}^{M} J_m^{altitude} \\ &+ w_4 \sum_{m=1}^{M} J_m^{Radar} + J_{threat} \\ J_{threat} =& \sum_{m=1}^{M} J_m^{Terrain} + \sum_{m=1}^{M} J_m^{Obstacle} + J^{Coor} \end{aligned} \tag{10}$$

where w_1, w_2, w_3 and w_4 are the weight of path length cost, smoothing cost, altitude cost and radar threat cost, respectively.

3 The basic PSO and SOS algorithm

In the path planning problem, for the case of multiple UAVs, in order to solve the coordination problem, a multi-UAV path planning model based on TSS technology mentioned above is used. The model simplifies the processing of multi-UAV coordination cost, and determines that the variables to be optimized are the height change between adjacent path points and the Y-direction velocity of each path point in the local coordinate system. Considering that the path of the UAV is obtained by connecting a series of path points, the entire path can be obtained by finding the coordinates of these path points. In this paper, a new hybrid optimization algorithm is proposed to solve the optimal solution search problem of the TSS model.

3.1 Review to the basic PSO

Particle swarm optimization (PSO) is a biologically inspired evolutionary optimization algorithm based on a random population. The idea of the algorithm is to initialize a group of particles in the solution space, in which each particle represents the candidate solution of the optimization problem to be solved. The characteristics of particle is represented by the position, speed and fitness value. The speed of the particle is affected by inertia, local optimum, and global optimum. During the iteration, the speed and position of the particles are constantly updated

$$\begin{aligned} v_{i,k+1} &= w v_{i,k} + c_1 r_1 (p_{best}^i - x_{i,k}) + c_2 r_2 (p_{gbest} - x_{i,k}) \\ x_{i,k+1} &= x_{i,k} + v_{i,k+1} \end{aligned} \tag{11}$$

where $v_{i,k}$ and $x_{i,k}$ are the speed and position of the ith particle at the kth iteration, respectively; w, c_1 and c_2 are the inertia weight, cognitive coefficient and social coefficient, respectively; p_{best}^i and p_{gbest} are the local optimum of particle i and global optimum, respectively; r_1 and r_2 are the random values within the range of [0,1].

At the same time, the fitness value of each particle is calculated, and the local optimum and global optimum are updated. After continuous iteration, it is finally expected to converge to the optimal solution within the specified number of iterations. The flow chart of basic PSO is shown in Fig. 3

3.2 The SOS algorithm

The symbiotic organism search (SOS) algorithm is a swarm intelligence optimization method, in which the organisms in the population produce new offspring through symbiotic behavior. In SOS, there are three kinds of operations, namely mutualism operation, commensalism operation and parasitism operation. These three operations can well simulate the interaction model between organisms. The individual in the population generate new off-spring through these three operations, and the entire process makes the population evolve in a favorable direction.

Here we only introduce the commensalism operation in SOS, which is used by the proposed method. The commensalism refers to the interaction between two organisms, which has no effect on one side and is beneficial to the other side. It is a "positive interaction" relationship. For organism O_i, randomly choose organism O_j from the population to form a commensalism relationship, where O_i benefits from this relationship, but O_j is not. Organism O_i updates as follows

$$O_{inew} = O_i + r\left(O_{best} - O_j\right) \tag{12}$$

where r is a random parameter in [-1, 1] and O_{best} represents the best organism in the population.

4 The development of the HIPSO-MSOS algorithm

4.1 Basic idea of the proposed algorithm

In order to improve the quality of the solution when solving multi-objective optimization problems, most intelligent optimization algorithms are committed to finding the most effective balance between global exploration capability and local exploitation capability. In PSO, individuals are updated based on inertia, local optimum, and global optimum. At the same time, the exploration and exploitation capability depend mainly on the parameters w, r_1 and r_2.

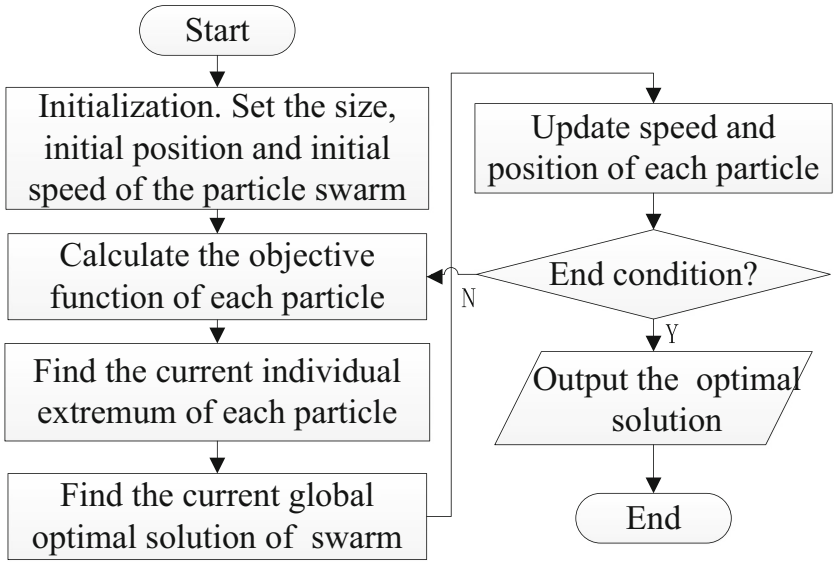


Fig. 3 The implementation flowchart of basic PSO

When the value of parameter w is large and the parameter r_i is small, the particle performs global exploration. Conversely, when the value of parameter w is small and the parameter r_i is large, it means that a local exploitation is required. In SOS, when an individual attempts to update, it will randomly select another individual from the current population and combine it with the current optimal individual. Compared with PSO, SOS algorithm has better local exploitation capability. Therefore, in order to better solve the UAV path planning problem in complex environments, we have improved the PSO algorithm to retain the exploration capability and improve the

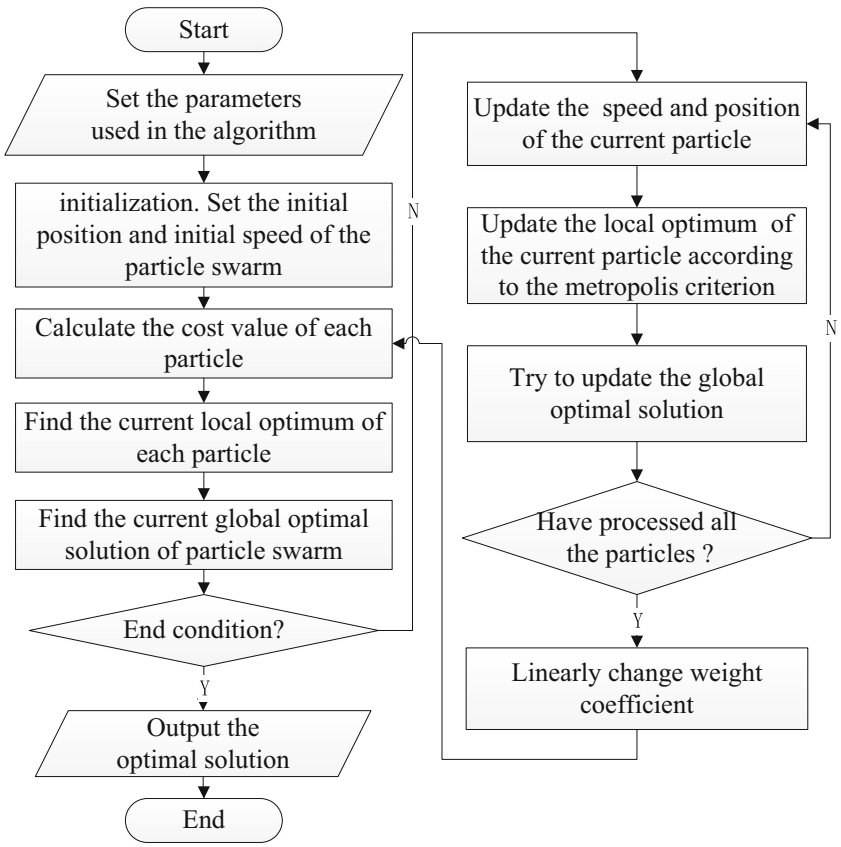


Fig. 4 The flow chart of improved PSO

convergence accuracy. Then, the commensalism operation of the SOS algorithm is modified to enhance its exploitation capability. Finally, combining the advantages of improved PSO (IPSO) and modified SOS (MSOS), a novel hybrid algorithm HIPSO-MSOS is proposed.

4.2 The improved PSO algorithm

Because the PSO has the disadvantages of easy premature convergence (especially in complex multi-peak search problems) and poor local optimization performance, the following improvements are adopted to PSO.

(1) The metropolis criterion is a method to determine whether to accept the new state, which can be used to improve the situation that PSO algorithm falls into local optimum prematurely. Therefore, the metropolis criterion is adopted as the acceptance strategy for the updated solution of PSO:

1) Calculate the fitness value of $x_{i,k}$ (the position of particle i at the kth iteration) according to the cost function.
2) If the fitness of $x_{i,k}$ is better than the fitness of p_{best}^i (the local optimum of particle i), replace p_{best}^i with $x_{i,k}$.
3) Otherwise, if $rand(0,1) < exp(\Delta/T)$, replace p_{best}^i with $x_{i,k}$, where $\Delta = fitness(x_{i,k}) - fitness(p_{best}^i)$ and T is current temperature parameter.

(2) Linear change weight coefficient. The basic idea is to dynamically change the weighting coefficients of inertia, cognitive component and social component, so that the search strategy is affected by different factors at different stages.

$$\begin{aligned} w &= w_{max} - (w_{max} - w_{min}) \big/ (w_{max} - w_{min}) Iter_{Max}\ Iter_{Max} \times iter \\ c_1 &= c_{max} - (c_{max} - c_{min}) \big/ (c_{max} - c_{min}) Iter_{Max}\ Iter_{Max} \times iter \\ c_2 &= c_{min} + (c_{max} - c_{min}) \big/ (c_{max} - c_{min}) Iter_{Max}\ Iter_{Max} \times iter \end{aligned} \tag{13}$$

where w_{min} and w_{max} are the minimum and maximum weight coefficients of inertia, respectively; c_{min} and c_{max} are the minimum and maximum weight coefficients of the cognitive component and social component, respectively; $Iter_{Max}$ and $iter$ are the maximum iterations and current iterations, respectively.

It can be known from (13) that the inertia weight w and the cognitive coefficient c_1 decrease linearly with the increase of the number of iterations, and the social coefficient c_2 increases linearly with the increase of the number of iterations. Therefore, it has strong global search ability in the early stage of algorithm, and has strong local convergence ability in the later stage, which improves the performance of the algorithm. The flow chart of improved PSO is shown in Fig. 4.

(3) Update the optimal solution information in real time. After each particle has updated its position, it will be compared with the current optimal solution. If it is better than the optimal solution, it will immediately replace the current optimal solution, so that other particles that have not been updated can obtain the latest optimal solution. The pseudo-code of improved PSO is defined as follows:

Algorithm 1 The IPSO pseudocode.

Initialize:

1. Set the parameters, calculate the population initialization range;
2. Initialize particle position, velocity, and p_{best}^i;
3. Calculate the cost value of each particle;
4. Evaluate g_{best}

Optimize:

5: **for** $k = 1$ to $Iter_{Max}$
6: **for** $i = 1$ to N
7: Update the Velocity $v_{i,k}$ and position $x_{i,k}$ using (11);
8: Calculate the cost value of ith particle $J_{cost}(x_{i,k})$
9: **if** $J_{cost}(x_{i,k}) < J_{cost}(g_{best})$
10: Update g_{best} with $x_{i,k}$
11: **end if**
12: $\Delta = J_{cost}(x_{i,k}) - J_{cost}(p_{best}^i)$
13: **if** $\Delta < 0 || rand(0,1) < exp(-\Delta/T)$ **then**
14: Update p_{best}^i with $x_{i,k}$
15: **end if**
16: **end for**
17: Update the weight of inertia w,cognitive coefficient c_1 and social coefficient c_2 by (13)
18: **end for**
19: **Return** g_{best} **as optimal solution**
20: **Terminate**

4.3 The HIPSO-MSOS algorithm

During each iteration, IPSO is used firstly to update each individual in the population. At this stage, individuals are affected by the inertia component, cognitive component, and social component to update their positions. During the iteration process, the coefficients of these three components change linearly, which makes the algorithm perform global

exploration in the early stage and local exploitation in the later stage. Then, better results are obtained through the modified commensalism operation. At this stage, individuals try to update their position based on the current optimal individual and a randomly selected individual from the population. If the updated individual is better, the update is successful; otherwise the update fails and the individual's position will not change. The algorithm performs the next iteration until the termination condition is met. The steps of the proposed algorithm are as follows:

Step 1: Initialization

In this step, the particle swarm size N and the maximum number of iterations $Iter_{max}$ are set. As mentioned in Section 2.2, each particle represents the velocity component and height increment of each UAV at each waypoint. Meanwhile, the range of velocity components and height increments can be calculated from the UAV's maximum yaw angle ψ_{max}, and maximum pitch angle ϕ_{max}. Therefore, the initial range of the particle swarm can be limited to make the search process more efficient.

Step 2: Updating individual position with IPSO

The position of each particle is updated according to (11), and the inertia weight, cognition coefficient, and social coefficient are changed according to (13). In addition, the metropolis criterion is used to determine whether the local optimum of particle needs to be updated. At the same time, the optimal solution information is updated in real time.

Step 3: Updating individual position with MSOS

During the algorithm iterative process, the commensalism operation in SOS is conducive to the exploitation of the local optimum. In order to improve the exploitation ability of the proposed hybrid algorithm, the modified commensalism operation in SOS is introduced. At this operation, an individual O_j is randomly selected from the population and combine the current optimal individual O_{best} to try to update the individual O_i. The new offspring of O_i are influenced by O_{best} and O_j. In addition, if the fitness value of the new offspring is not as good as the original individual, the update fails. O_j performs the same operation at the same time. In the modified commensalism operation, individuals update according to:

$$\begin{aligned} O_{inew} &= O_i + r\left(O_{best} - O_j\right) \\ O_{jnew} &= O_j + r\left(O_{best} - O_i\right) \end{aligned} \tag{14}$$

where r is a random parameter in [-1, 1] and O_{best} represents the best individual in the population, which is equivalent to the global optimal solution P_{gbest} in PSO. If O_{inew} is better than O_i, replace O_i with O_{inew}, otherwise do nothing, and the same applies to O_j.

Step 4: Go to step 2 until the termination condition is met

The algorithm will run until the maximum number of iterations is reached or the optimal solution for the optimization problem is found. The pseudo-code of the proposed algorithm is defined as follows:

Algorithm 2 The HIPSO-MSOS pseudocode.

Initialize:

Set the parameters, calculate the population initialization range;
Initialize particle position, velocity, and p^i_{best};
Calculate the cost value of each particle;
Evaluate g_{best}

Optimize:

for $k = 1$ to $Iter_{Max}$ **do**
  for $i = 1$ to N **do**
    Update the Velocity $v_{i,k}$ and position $x_{i,k}$ using (11);
    Calculate the cost value of ith particle $J_{cost}(x_{i,k})$
    Calculate $x_{inew,k}$ and $x_{jnew,k}$ by (14) (x_j is selected randomly)
    if $J_{cost}(x_{inew,k}) < J_{cost}(x_{i,k})$ **then**
      Update $x_{i,k}$ with $x_{inew,k}$
    end if
    if $J_{cost}(x_{jnew,k}) < J_{cost}(x_{j,k})$ **then**
      Update $x_{j,k}$ with $x_{jnew,k}$
    end if
    if $J_{cost}(x_{i,k}) < J_{cost}(g_{best})$ **then**
      Update g_{best} with $x_{i,k}$
    end if
    $\Delta = J_{cost}(x_{i,k}) - J_{cost}(p^i_{best})$
    if $\Delta < 0 || rand(0, 1) < exp(-\Delta / T)$ **then**
      Update p^i_{best} with $x_{i,k}$
    end if
  end for
  Update the weight of inertia w,cognitive coefficient c_1 and social coefficient c_2 by (13)
end for
Return g_{best} **as optimal solution**
Terminate

5 Path smoothing based on B-spline curve

Because the proposed algorithm produces a series of path points, which cannot meet the actual flight requirements of UAV, further smoothing is needed. The B-spline curve can generate a smooth path with continuous curvature by taking the series of path points as the control points of B-spline basis function. Suppose that the path point sequence

Table 1 The parameters of UAV

maximum yaw angle	45°	maximum pitch angle	45°
maximum flight height	300m	minimum flight height	50m
maximum flight speed	200m/s	minimum flight speed	80m/s

of a feasible path is $\{p_1, \cdots, p_n\}$. In this paper, the cubic B-spline curve is used, and its basis function is

$$\begin{cases} G_{0,3} = \frac{1}{6}(-t^3 + 3t^2 - 3t + 1) \\ G_{1,3} = \frac{1}{6}(-3t^3 - 6t^2 + 4) \\ G_{2,3} = \frac{1}{6}(-3t^3 + 3t^2 + 3t + 1) \\ G_{3,3} = \frac{1}{6}t^3 \end{cases}, t \in [0, 1] \quad (15)$$

The cubic B-spline curve segment $p_{i,i+3}$ between the path point p_i and the path point $p_{(i+3)}$ is

$$p_{i,i+3} = \frac{1}{6}\begin{bmatrix} 1 & t & t^2 & t^3 \end{bmatrix}\begin{bmatrix} 1 & 4 & 1 & 0 \\ -3 & 0 & 3 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}\begin{bmatrix} p_i \\ p_{i+1} \\ p_{i+2} \\ p_{i+3} \end{bmatrix}, t \in [0, 1] \quad (16)$$

After smoothing, the resulting path satisfies the performance constraints of UAV, which is flyable for UAV.

6 Simulation experiment

Based on the above algorithm design, the simulation experiment was performed using jMonkeyEngine programming on an ordinary PC with Intel 2.5GHZ and 8GB memory. Using jMonkeyEngine to build a three-dimensional virtual task environment is convenient for us to visualize the planned path in the task environment. In this section, in order to evaluate the effectiveness of the HIPSO-MSOS algorithm, simulation experiments were performed respectively in two fields. One of them include obstacles and radar (case1), and the other included terrain additionally (case2).

Table 2 The starting and target point of UAVs

No.	Start(m)	Target(m)
UAV1	(1200,1700,0)	(1200,-1200,50)
UAV2	(1200,1700,0)	(0,-1200,50)
UAV3	(-800,1700,0)	(0,-1200,50)
UAV4	(0,1700,0)	(-1200,-1200,50)

In the simulation experiment, the planned area was set to 4km * 4km*0.5km, and an attempt was made to plan the optimal safe path from the starting point to the target point for 4 UAVs. The UAV parameter settings are shown in Table 1. The starting point and target point corresponding to each UAV are shown in Table 2.

The results are averaged 30 independent runs. In the planning field, the radar is represented by two hemispheres, and the obstacles are represented by seven cuboids, as shown in Table 3. The HIPSO-MSOS algorithm parameters are listed in Table 4. The comparative results with the PSO, FireFly, DE, MSOS and HSGWO-MSOS algorithms are also given in Table 5.

For the first case, Fig. 5 shows the experimental results of the proposed algorithm without terrain. We can see that the results of HIPSO-MSOS algorithm planning can meet the requirements of path planning. The planned path is short in length, low in height and relatively smooth. When UAV flies along the path, it can not only avoid obstacles and radar detection area but also not collide with colleagues.

For the second case, we perform simulation experiments in a complex environment including terrain. The path planning results of the proposed algorithm are shown in Fig. 6. As can be seen from the figure, the path planned by the HIPSO-MSOS algorithm for each UAV can meet the predetermined requirements. The convergence curves of the six algorithms in case2 are illustrated in Fig. 7. Obviously, the convergence effect of the proposed algorithm is better than the other five algorithms. HIPSO-MSOS is superior to DE algorithm and FireFly algorithm both in convergence speed and convergence accuracy. The convergence speed of HIPSO-MSOS is not as good as PSO due to the negative influence of SOS algorithm, but the convergence accuracy is better than PSO. However, the MSOS algorithm was caught in the local optimization prematurely, so that the cost of the optimal solution eventually converged is large. The convergence speed of HSGWO-MSOS algorithm is slowest, and the convergence accuracy is worse than HIPSO-MSOS.

Table 3 The information of the obstacles and radars

Threat	Threat center(m)	Size/Radiu(m)
obstacles	(1200,800,0)	(140,140,120)
	(-1000,-800,0)	(180,180,130)
	(0,0,0)	(200,200,150)
	(1200,-800,0)	(300,200,150)
	(-1200,800,0)	(300,200,150)
	(200,800,0)	(300,200,150)
	(-200,-800,0)	(300,200,150)
radars	(900,50,0)	600
	(-900,0,0)	600

Table 4 Parameters involved in the HIPSO-MSOS

Population size	50	Maximum iteration	300
Waypoints number	20	Maximum inertia weight	1
Minimum inertia weight	0.4	Maximum social coefficient	2.5
Minimum social coefficient	0.5	Maximum cognitive coefficient	2.5
Minimum cognitive coefficient	0.5		

Table 5 The comparison results of the four algorithms

	PSO	FireFly	DE	MSOS	HSGWO-MSOS	HIPSO-MSOS
best	3609.95	3013.25	3608.41	9285.03	2906.59	3080.16
worst	5701.57	7490.31	5613.25	14388.30	7442.76	4290.43
mean	4292.45	5177.67	4052.12	12022.01	4419.28	3442.54
std	708.86	1441.04	414.87	1329.74	1070.89	315.80

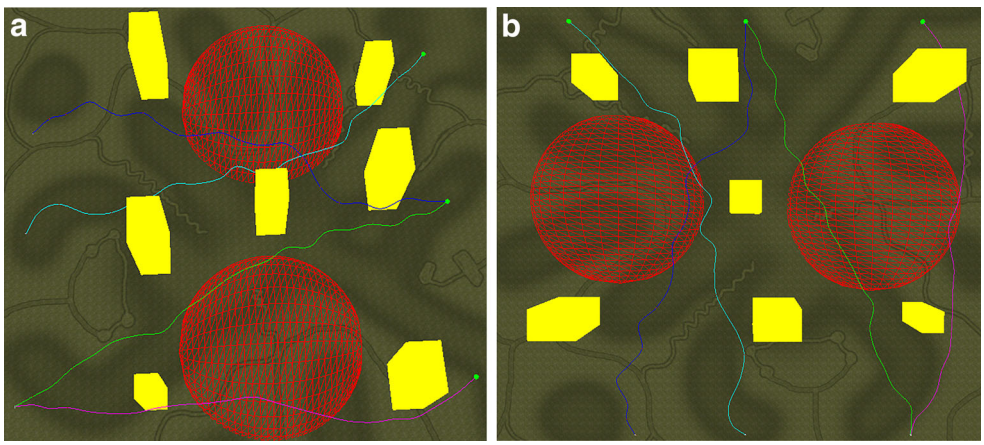


Fig. 5 The planning result of HIPSO-MSOS in case1

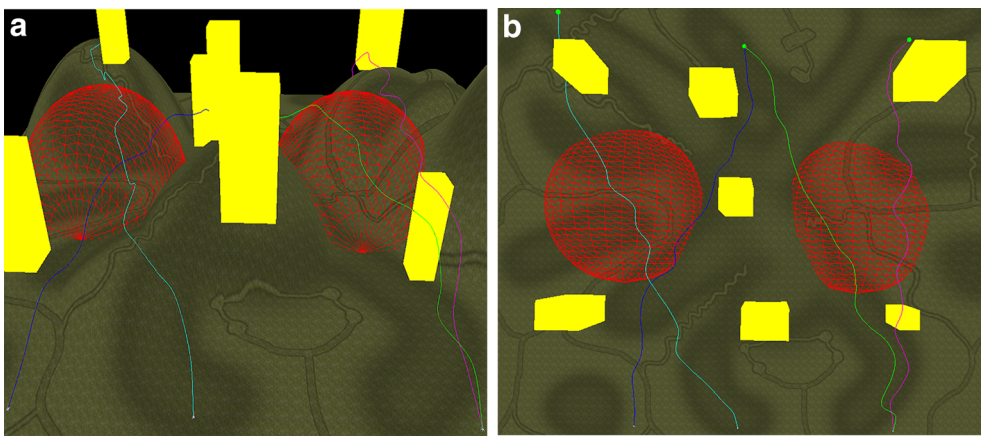


Fig. 6 The planning result of HIPSO-MSOS in case2

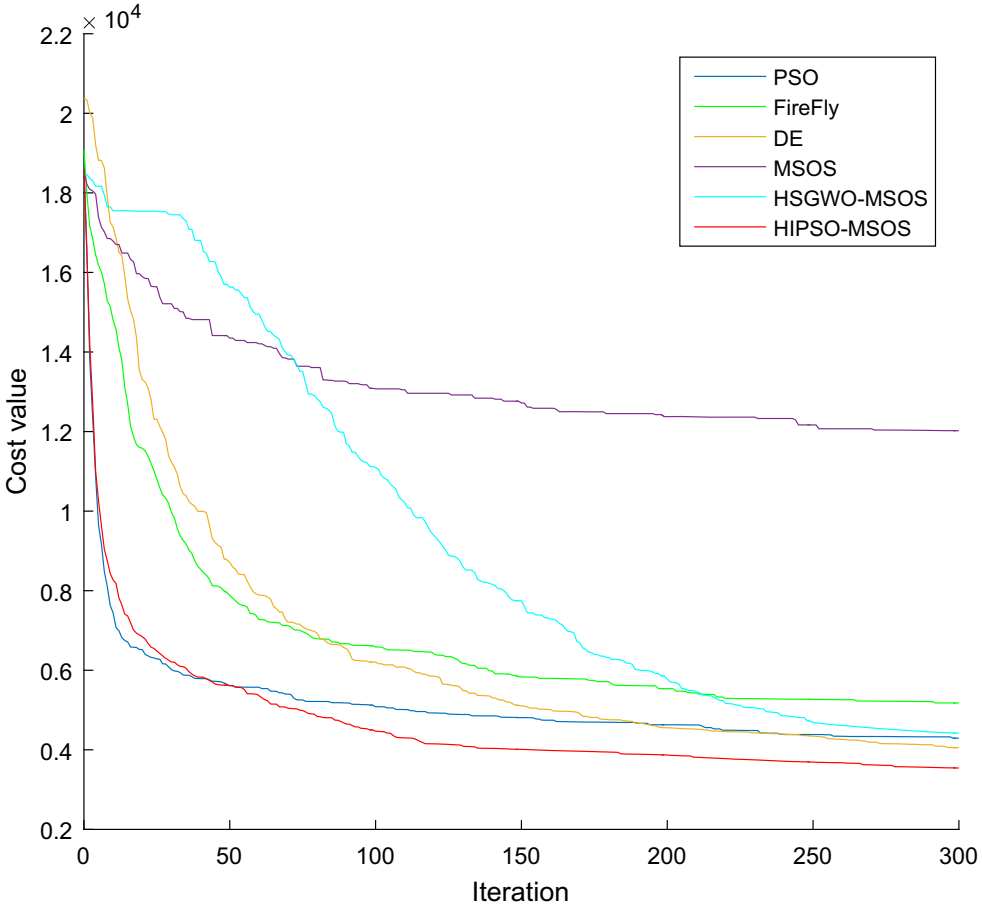


Fig. 7 The convergence curves of the six algorithms in case2

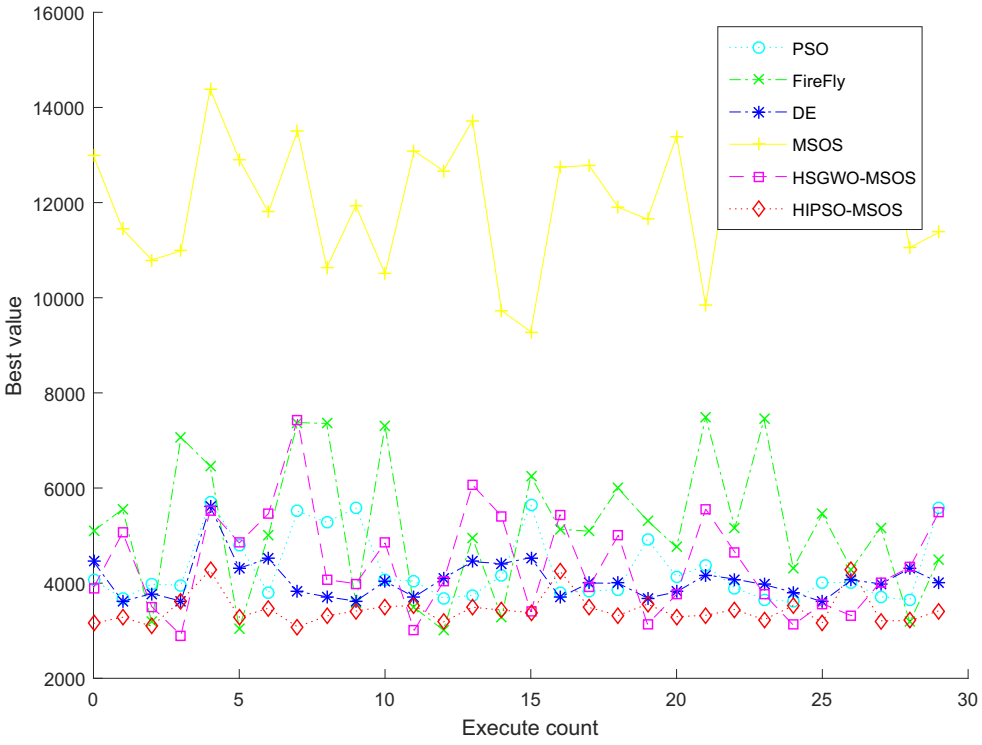


Fig. 8 The line chart of six algorithms in case2

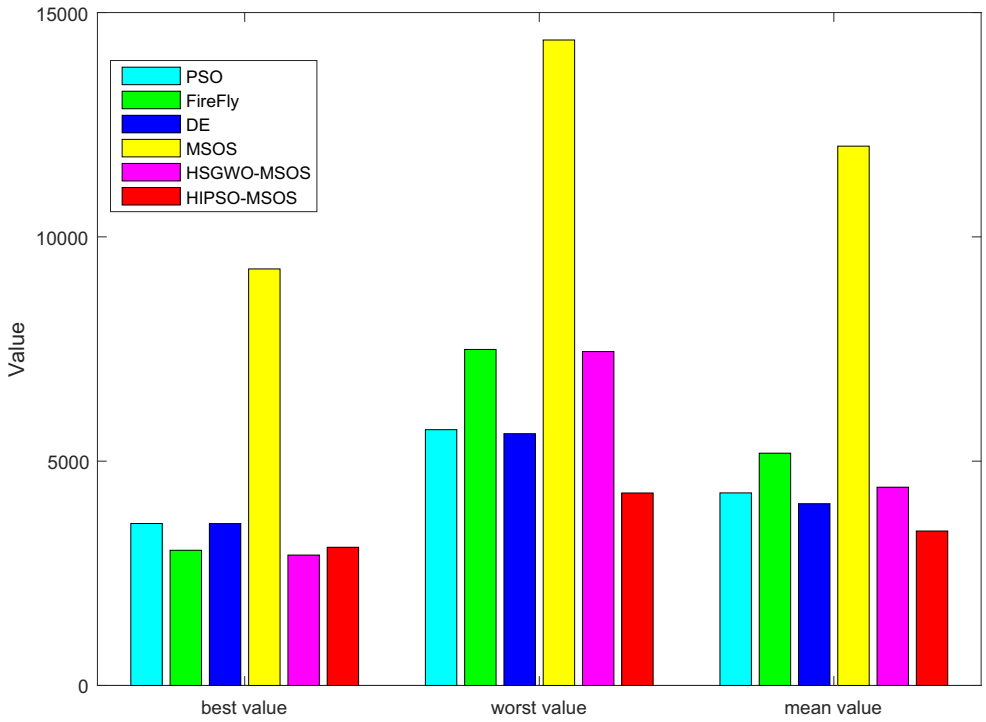


Fig. 9 The statistical results of the six algorithms in case2

The line chart composed of the optimal solutions obtained by each algorithm in 30 independent experiments is shown in Fig. 8. The statistical results are illustrate in Fig. 9 and Table 5. The MSOS has the worst performance on all value. Although HMGWO-MSOS has the smallest best value, its mean and standard deviation values are larger than proposed algorithm, which means its poor performance. DE is superior to PSO, FireFly, HMGWO-MSOS and MSOS in mean and standard deviation values, but significantly worse than HIPSO-MSOS. The simulation results of the proposed algorithm are lower than the other three algorithms in terms of worst value, mean value and standard deviation value, and the lower standard deviation value proves that HIPSO-MSOS can stably search for the best path.

In summary, it is almost impossible to obtain a feasible collision-free path using MSOS alone. Although PSO, DE, FireFly and HSGWO-MSOS can sometimes obtain feasible paths, it is not satisfactory in terms of path length, smoothness, etc. However, due to the combination of the advantages of the two algorithms, the proposed algorithm HIPSO-MSOS has good performance in all aspects, and can plan the safe, cooperative, collision-free optimal path for each UAV.

7 Conclusions

This paper solves the problem of coordinated path planning for multiple UAV in complex three-dimensional environment. In the method proposed in this paper, under the premise of considering performance constraints of UAV, the TSS model is introduced to simplify the calculation of the coordination cost between the UAVs. In order to make the planned path meet the predetermined target performance, a reasonable multi-objective optimization function is designed. Next, a novel hybrid particle swarm optimization algorithm named HIPSO-MSOS is proposed to solve optimization problem. Then, smoothing the generated paths by the cubic B-spline curve to make it traceable to UAV. Finally, the experimental results show that the HIPSO-MSOS algorithm can successfully generate an effective, safe and cooperative path for each UAV, and the comparison results also show that the HIPSO-MSOS algorithm has higher efficiency optimization capabilities than other algorithms.

In our proposed algorithm, in order to enhance the search efficiency of the algorithm, some effective operators are integrated. The metropolis criterion is used to prevent the PSO algorithm from falling into local optima prematurely, linear change strategy of the weight coefficient makes the particles affected by different components at different stages, which greatly enhances the convergence performance of the algorithm and the instant update strategy of the optimal solution ensures the real-time performance of the optimal solution to improve the convergence accuracy of the algorithm. In addition, by combining the commensalism operation in the SOS algorithm, the local search capability of the algorithm can be effectively enhanced.

The simulation results confirm that HIPSO-MSOS has a significant advantage in terms of accuracy, convergence speed, stability. This study sheds light on thought for further improving the performance of PSO algorithm.

At present, the method we proposed is only applicable to the situation of multi-UAV flying in a fixed formation. It is difficult to apply to dynamic mission scenarios and the UAV cannot avoid collision with dynamic obstacles in the environment when flying along the planned path. Therefore, future research will focus on formation control and real-time obstacle avoidance.

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